

Self-testing questions - miscellaneous

Heat treatment

Reasons for performing surface hardening of steels

Suitable initial microstructure for heat treatment of hypo- and hyper-eutectoid steels

Deleterious alloying element in heat treatment of steel with reference to residual austenite

Consequence of residual austenite after heat treatment of carbon steels

In chemical surface treatment, how does penetration depth correlate with superficial residual stress

Among the available surface hardening processes, which is the most suitable to withstand point-load crushing

The most low-cost effective surface hardening process

The surface hardening allowing the deepest hardened depth

The two alloying elements promoting the deepest penetration depth of martensite

The most cost-effective surface hardening process combining hardness, penetration depth, satisfactory compressive superficial residual stresses

The most durable superficial case-hardened layer for gears

Is hardenability specially requested for a large size rotating-bending shaft ?

TTT curves

Which are the two basic metallurgical phenomena intrinsic in the initial TTT curve?

What does the inflection point of a sigmoidal growth kinetics represent?

What is the basic feature of the nose temperature of a TTT curve?

How much small (or large) shall be both nucleation and growth rates to produce a fine microstructure after non-equilibrium isothermal treatment?

How much small (or large) shall be the requested undercooling to produce a fine microstructure after non-equilibrium isothermal heat treatment?

What are the main functions of austenitization prior the heat treatment of steels?

How do nucleation and growth phenomena change when passing from isothermal heat treatment to continuous heat treatment?

What is the main external factor affecting the growth step of an isothermal phase transformation?

What are the main factors affecting the nucleation step of an isothermal phase transformation?

Recovery and Recrystallization (provide a **separate answer** to the following quests)

What are the main metallurgical factors involved in the recovery/recrystallization of a metal being heated after cold work?

How do the mechanical properties of a metal change after recovery/recrystallization?

What are the main recovered properties in a metal after recovery/recrystallization?

What is the main factor affected by recovery/recrystallization?

How does the distorted-grain microstructure is changed after recovery/recrystallization?

How does the dislocation density is changed during recovery/recrystallization

How does the microstructure change in the limit of long time recrystallization

What are the fundamental metallurgical mechanisms governing the recrystallization process

Upon comparing the metallurgical kinetics governing recrystallization and phase transformations, what are the main differences (if any)?

Why after recrystallization of a metal its grains are so small and soft?

Upon comparing large and small grain sizes of a metal, which of them does retain largest ductility

Damascus Swords

Which are the microstructure features in Damascus steels maximizing its:

- fatigue resistance
- wear resistance
- toughness
- strength
- corrosion resistance
- bending strength
- impact strength
- cutting edge

Among the studied technological processes, which is the one(s) most resembling that employed by old blacksmith?

What are the main strengthening mechanism exploited in the manufacturing of the sword? Provide a plausible explanation of each.

What are the hard phases?

Do alloy elements contribute to the strength/toughness; if yes, how have they been introduced?

Strengthening and Microplasticity

How a microstructure shall be to maximize its strength/toughness combination?

Upon comparing the effects of LAGB and HAGB, which of them maximizes ductility (if any) and which maximizes strength (if any)?

In strain hardening of metals which is the lattice defect leading the main role?

In strengthening of metals, which is the main common factor to all methods?

Does overaging lead to enhanced ductility, strength, brittleness? And why?

In precipitation hardening how strength and particle size will change if:

- the prior cooling is not a severe quenching?
- The solutionizing temperature is lower than expected?
- Precipitation mainly takes place at grain boundaries rather than inside the grain?
- The temperature of artificial aging is above the solutionizing temperature?
- The quenching is performed at sub-zero temperature?

In the microplasticity of metals, under an imposed stress over:

- two grains by a normal stress, which grain locations are first involved by dislocation activity?
- two grains by a normal stress, which grain locations are likely to fail due to overloading
- a polycrystalline material by shear, which grains are first involved in the dislocation activity?

If an alloy attains maximum strengthening by plastic deformation, what kind of fracture behaviour will undergo?

In which condition plastic deformation by twinning is favoured versus dislocation gliding?

In which crystal structures is twinning favoured?

How does the stacking fault energy affect twinning?

To promote formability by plastic deformation of a material, how shall be the strain hardening coefficient?

Which are the characteristic values of the strain hardening coefficient for FCC and BCC?

Among the BCC and FCC crystals, which of both can strain harden to maximum limit?

Among ferrite and austenite in the respective stainless steels, which of them can strain harden to larger limit?

In which materials twinning is the dominant plastic deformation process?

What is the main peculiar feature of twinning deformation?

As a rule of thumb, given 100% the supplied mechanical energy, what is the percentage utilized for plastic deformation and what is lost by heat dissipation?

Some tensile sample exhibits in combination of a ductile fracture also a brittle fracture at its core after testing, why?

Which engineering alloys do exhibit the largest strain hardening coefficient? Why?

What is the constitutive law governing plastic deformation under tensile test?

Why grain refinement is the only strengthening mechanism allowing strengthening without penalizing toughness?

What are the materials processing strategies to combat fatigue failure?

What is the current technological process performed industrially to prevent Luders' band ?

What is the current technological process performed industrially to promote dent resistance of panels?

What are the Cottrell atmospheres? and their inherent practical consequences?

